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MetaCommunityMetrics.jl: A Julia Package for Spatiotemporal Metacommunity Analysis

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Software

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Summary

Community ecologists are often interested in infering ecological processes from observational data. This involves calculating different metrics such as beta diversity. Calculating these metrics is easy and fast due to the availability of R packages. However, community ecology today is growing beyond analysing single datasets. Large biodiversity databases have expanded the sets of spatio-temporal community data, and there is a growing interest in using simulations to study ecological processes. For these applications, R packages become slow to use.

`MetaCommunityMetrics.jl` is a Julia package that helps ecologists to analyze large datasets of species observations to understand how community composition varies across space and time. By implementing computationally optimized versions of established software and new complementary functions that are designed for handling large spatiotemporal datasets, this package enables ecologists to efficiently analyze large-scale spatiotemporal community datasets that would be computationally prohibitive with existing tools. We designed intuitive interfaces that would feel familiar to ecologists transitioning from R. Our function naming conventions closely match their R equivalents, and we maintained parameter naming and ordering consistency where possible. Some of the metrics we have implemented include beta diversity decompositions in space and time, dispersal-niche continuum index, and variability metric, but a full list of metrics is listed in Table 1. Full documentation and examples, including installation instructions and help files, are available at https://cralibe.github.io/MetaCommunityMetrics.jl/.

Table 1: A summary of the metrics and functions available in `MetaCommunityMetrics.jl`.

Metrics	Functions	Details
Beta diversity decompositions in space and time	`beta_diversity()`	Re-implementation of R package `adespatial`(Dray et al., 2023)
	`spatial_beta_div()`	Re-implementation of R script from Guzman et al. (2022)
	`temporal_beta_div()`	Re-implementation of R script from Guzman et al. (2022)
Dispersal-niche continuum index (DNCI)	`create_groups()`	A novel grouping function for DNCI analysis
	`plot_groups()`	A novel visualization function for DNCI analysis
	`DNCI_multigroup()`	Re-implementation of R script from R package `DNCImper` (Gibert, 2024)
Niche overlap index	`niche_overlap()`	Novel implementation
Occupied patches proportion	`prop_patche()`	Re-implementation of R script from Guzman et al. (2022)

Metrics	Functions	Details
Variability Metric	`CV_meta()`	Re-implementation of R script from Wang et al. (2019)
Niche hypervolume measurements	`MVNH_det()`	Re-implementation from R package `MVNH`(Lu et al., 2021)
	`MVNH_dissimilarity()`	Re-implementation from R package `MVNH`(Lu et al., 2021)
	`average_MVNH_det()`	Novel extension of `MVNH_det()`
	`average_MVNH_dissimilarity()`	Novel extension of `MVNH_dissimilarity()`

Statement of need

Many valuable R (R Core Team, 2023) packages have been developed to aid in community analyses by providing ecological metrics that can summarize processes from biodiversity patterns and have been widely adopted in the field, such as `adespatial` (Dray et al., 2023), `codyn` (Hallett et al., 2020), and `vegan` (Oksanen et al., 2024). However, conducting community analyses in R is often computationally expensive, especially when working with large datasets, a limitation that becomes critical in workflows requiring repeated metric calculations. These include simulation studies for method validation, bootstrap resampling for uncertainty estimation (Efron, 1992), null model approaches for statistical testing (Gotelli & Graves, 1996), and the analysis of large biodiversity databses, such as BioTime, all of which can create significant computational bottlenecks when processing multiple large datasets iteratively. To improve the efficiency of these computationally intensive analyses, `Julia` becomes a logical choice for re-implementations. `Julia` is known for its just-in-time compilation that optimizes code execution to levels comparable with lower-level languages like C or Fortran (Bezanson et al., 2017), yet Julia remains relatively uncommon in ecological workflows. During development, we conducted a comprehensive review of existing `Julia` packages under the EcoJulia organization, a community-driven effort providing tools for ecological and environmental analysis, and the PoisotLab, which develop many `Julia` packages focusing on quantitative and computational ecology. We ensure that our package builds beyond the tools currently existing in `Julia` even if they share similar ecological concepts. For example, our beta diversity metrics share a similar concept, beta diversity, with `Diversity.jl` (Robert Reeve & Harris, 2024). However, our implementation decomposes beta diversity into species replacement and richness differences (or abundance difference when abundance data is used instead of occurrence data) (Baselga, 2010, 2012, 2013; Legendre, 2014), which differentiates our approach from `Diversity.jl`, which emphasizes alpha, beta, and gamma diversity measures at the metacommunity and community levels (Richard Reeve et al., 2014). To our knowledge, no existing Julia packages implement the specific metrics listed in Table 1.

`MetaCommunityMetrics.jl` delivers substantial computational advantages over existing tools. For example, we re-implemented the `beta.div.comp` function from the R package `adespatial` (Dray et al., 2023), achieving up to $\sim 58\times$ faster execution times while maintaining methodological consistency (see Figure 1). As shown in Figure 1, performance gains range from $\sim 3\times$ to $58\times$ across all re-implemented functions using three testing datasets (large: 53,352 observations; medium: 26,676 observations; small: 5,325 observations), positioning `MetaCommunityMetrics.jl` as an efficient solution for diverse ecological research needs, from standard biodiversity analyses to computationally intensive simulation studies and large-scale data processing.

Beyond performance improvements, we extended pre-existing R implementations to handle spatiotemporal datasets directly. We added temporal, spatial, and community-level aggregation methods to metrics originally restricted to a single time point, site, and species pair, eliminating extensive data preprocessing. For instance, beta diversity decompositions in R operate on

a single temporal snapshot of a metacommunity, while niche hypervolume measurements calculate volumes for an individual species and dissimilarity between a single species pair. We extended these with `spatial_beta_div()` and `temporal_beta_div()`, which calculate beta diversity decompositions across time and space, respectively. Similarly, `average_MVNH_det()` calculates average niche volume across all species, and `average_MVNH_dissimilarity()` computes average niche dissimilarity across all species pairs. For DNCI calculations, we developed `create_groups()` and `plot_groups()` to meet the site grouping requirements suggested by Vilmi et al. (2021) and to visualize the resulting groupings.

Additional handling of edge cases when calculating DNCI

We extended our DNCI implementation to handle edge cases common in simulated community datasets (e.g., single-species communities, insufficient permutation variation) by returning status flags that identify when the standard DNCI calculation is not possible. See documentation for details on all five edge cases.

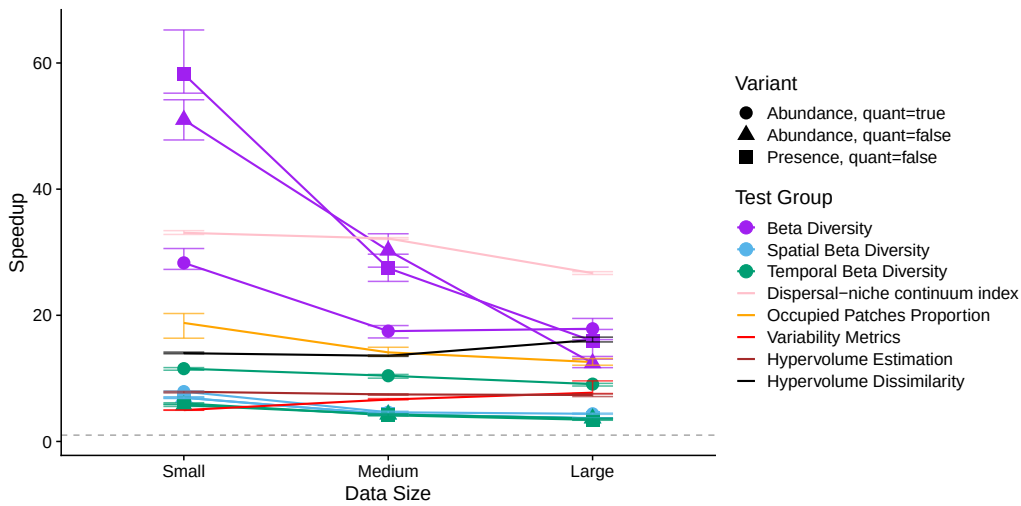

Figure 1: A plot showing the speedup of all benchmarked functions across three datasets: small (5,335 observations), medium (26,676 observations), and large (53,352 observations). Speedup is calculated as the `R` median execution time divided by the `Julia` median execution time. Median speedup and its confidence interval of each comparison is shown. The grey dashed line shows where speedup equals to 1, that is both `R` and `Julia`'s implementations require the same amount of time. The calcualation of beta diversity has two variants: `quant = true` (abundance data is used) and `quant = false`(occurrence data is used). Details about this benchmarking can be found in the documentation under the benchmark results section.

Validation

All re-implementations were validated against their R equivalents (see documentation).

Examples

Using rodent metacommunity data as the sample data

We demonstrate the package using rodent data from the Portal Project (Ernest et al., 2018), a long-term Chihuahuan desert ecosystem study. The sample dataset includes 21 species across 24 sites from 2010-2023 (117 sampling events). For the ease of demonstration, we simulated our own set of spatial coordinates and environmental variables (temperature, precipitation) for

the sample dataset. These data are used to demonstrate the functionailties of DNCI and niche hypervolume measurements. The data can be loaded with:

```
julia> load_sample_data()
```

The complete output of `load_sample_data()` is available in the documentation.

Demonstration using the sample data

Here, we demonstrate a typical community analysis workflow: (1) assessing community stability using variability metrics (`CV_meta()`), (2) decomposing beta diversity to understand spatial and temporal turnover (`spatial_beta_div()`, `temporal_beta_div()`), (3) inferring assembly processes via DNCI analysis (`create_groups()`, `plot_groups()`, `DNCI_multigroup()`), and (4) quantifying niche overlap patterns (`niche_overlap()`, `average_MVNH_det()`, `average_MVNH_dissimilarity()`).

Assessing temporal stability

We first examine how abundance variability differs across organizational levels (species vs. community) and spatial scales (local vs. regional) using `CV_meta()`:

```
julia> using MetaCommunityMetrics, Pipe, DataFrames, Random
julia> df = load_sample_data()
julia> CV_meta(df.Abundance, df.Sampling_date_order, df.plot,
         df.Species)
1×4 DataFrame
 Row  CV_s_l   CV_s_r   CV_c_l   CV_c_r
      Float64  Float64  Float64  Float64

   1  1.48859  0.944937  0.718266  0.580183
```

Variability in abundance decreases from local to regional scales (comparing `CV_s_l` to `CV_s_r` and `CV_c_l` to `CV_c_r`) and from species to community levels (comparing `CV_s_l` to `CV_c_l` and `CV_s_r` to `CV_c_r`), indicating greater stability at broader spatial scales and higher organizational levels.

Decomposing beta diversity

Next, we quantify compositional turnover and determine whether changes are driven by species replacement or abundance differences (`RichDif` equals to abundance differences when abundance data is used and equals to richness differeces when occurrence data is used) using `spatial_beta_div()` and `temporal_beta_div()`. The beta diversity decompositions in space:

```
julia> spatial_beta_div(df.Abundance, df.Sampling_date_order,
         df.plot, df.Species; quant = true)
1×3 DataFrame
 Row  spatial_BDtotal  spatial_Repl  spatial_RichDif
      Float64          Float64       Float64

   1         0.264822      0.121882         0.142939
```

The beta diversity decompositions in time:

```
julia> temporal_beta_div(df.Abundance, df.Sampling_date_order,
         df.plot, df.Species; quant = true)
1×3 DataFrame
 Row  temporal_BDtotal  temporal_Repl  temporal_RichDif
      Float64           Float64        Float64
```

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```
1          0.311222      0.0995483          0.211674
```

Spatial composition differences are driven equally by replacement and abundance differences (values are similar), while temporal differences are primarily driven by abundance changes (higher value), suggesting seasonal patterns or disturbance over time (Legendre, 2014; Podani & Schmera, 2011).

Inferring assembly processes

To determine whether communities are predominantly structured by dispersal or environmental filtering, we calculate DNCI using `DNCI_multigroup()`, which quantifies these relative contributions by comparing species-specific contributions to between-group dissimilarity in observed versus null model data. This function automatically filters absent or ubiquitous species and handles empty sites. Before calculating DNCI, we first group sites using `create_groups()` to ensure at least two groups with a minimum of five sites per group, and that variation in the number of species and sites per group does not exceed 40% and 30%, respectively:

```
julia> grouping_result = create_groups(df.Sampling_date_order,
         df.Latitude, df.Longitude, df.plot, df.Species, df.Presence)
```

`grouping_result` returns a dictionary indexed by `Sampling_date_order`. Time points not meeting grouping requirements are assigned `missing`. We can access the result at `Sampling_date_order` = 60 with `grouping_result[60]`.

Now, we will use `plot_groups()` to visualize the grouping at `Sampling_date_order()` = 60:

```
julia> plot_groups(grouping_result[60].Latitude,
          grouping_result[60].Longitude,
          grouping_result[60].Group; output_file = "groups.svg")
```

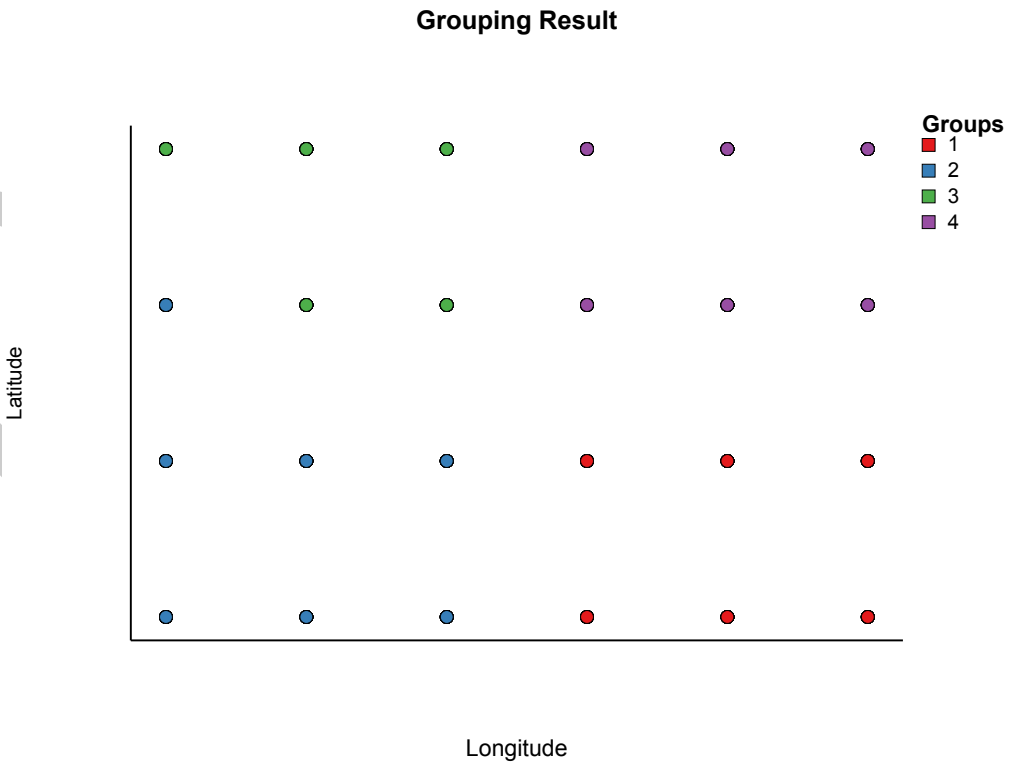

Figure 2: A plot showing the location of sites and their groups. Different colors represent different groups at `Sampling_date_order` = 60.

Figure 2 shows the grouping result at `Sampling_date_order` = 60.

We can then join `df` with `grouping_result[60]` to obtain the matrix and group assignment column for calculating DNCI at `Sampling_data_order` = 60 using `DNCI_multigroup()`.

```
julia> group_df = @pipe df |>
                  filter(row -> row[:Sampling_date_order] == 60, _) |>
                  select(_, [:plot, :Species, :Presence]) |>
                  innerjoin(_, grouping_result[60], on = [:plot => :Site,
                  :Species], makeunique = true)|>
                  select(_, [:plot, :Species, :Presence, :Group]) |>
                  unstack(_, :Species, :Presence, fill = 0)
julia> comm= @pipe group_df |>
             select(_, Not([:plot,:Group])) |>
             Matrix(_)
julia> Random.seed!(1234)
julia> DNCI_result = DNCI_multigroup(comm, group_df.Group, 1000;
          Nperm_count = false)
6×6 DataFrame
 Row  group1  group2  DNCI      CI_DNCI  S_DNCI    status
      Int64   Int64   Float64   Float64  Float64   String

   1       1       2  -3.41127  2.17348  1.08674   normal
   2       1       3  -2.44866  2.05951  1.02976   normal
   3       1       4  -2.3671   2.45697  1.22848   normal
   4       2       3  -2.65022  2.28931  1.14466   normal
   5       2       4  -3.0168   2.43496  1.21748   normal
   6       3       4  -1.83521  1.9589   0.979449  normal
```

Positive DNCI values suggest niche processes dominate community assembly, while negative DNCI values suggest dispersal limitation is more influential at a given time point. DNCI values that do not differ significantly from zero suggest equal contributions from both processes at a given time point. The DNCI values at `Sampling_date_order` = 60 do not differ significantly from zero for group pairs 1-4 and 3-4, while the rest are significantly smaller than zero. This indicates that most local communities in this metacommunity are dominated by dispersal processes, while both dispersal and niche processes contribute evenly to some local communities at `Sampling_date_order` = 60. To draw conclusions about the entire metacommunity over time, we suggest running this function for all available time points and averaging across all group pairs at all time points.

Quantifying niche overlap

Finally, we examine how species partition niche space by using `niche_overlap()`, `prop_patches()`, `average_MVNH_det()`, and `average_MVNH_dissimilarity()`. We will start with `niche_overlap()`:

```
julia> niche_overlap(df.Abundance, df.Species, df.plot,
          df.Sampling_date_order)
1×3 DataFrame
 Row  mean_niche_overlap_index  min_niche_overlap_index  max_niche_overlap_index
      Float64                   Float64                  Float64

   1                 0.0923816                      0.0                 0.406837
```

Then, we will use `prop_patches()`:

```
julia> prop_patches(df.Presence, df.Species, df.plot)
1×3 DataFrame
```

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```
 Row  mean_prop_patches  min_prop_patches  max_prop_patches
      Float64            Float64           Float64

   1           0.734649         0.0833333               1.0
```

We then calculate the average niche volume and average niche volume dissimilarity across all species using `average_MVNH_det()` and `average_MVNH_dissimilarity()`, respectively. Note that these functions, unlike the others, require environmental data. Users need to (1) standardized their environmental variables and transform them to normal distribution (if necessary); and (2) remove singletons before using these functions. In our sample dataset, temperature and precipitation have been standardized and checked for normality. The following demonstrates how we use these functions:

```
julia> df = @pipe load_sample_data() |>
                     groupby(_, :Species) |>
                     filter(row -> sum(row.Presence) > 1, _)|>
                     DataFrame(_)
julia> env_data = @pipe df |>
          select(_, [:standardized_temperature,
          :standardized_precipitation])
julia> average_MVNH_det(env_data, df.Presence, df.Species; var_names =
         ["Temperature", "Precipitation"])
1.2103765096417536
julia> average_MVNH_dissimilarity(env_data, df.Presence, df.Species;
         var_names = ["Temperature", "Precipitation"])
0.03059942936454443
```

High spatial overlap (`mean_prop_patches` $\sim$ 0.73) combined with low temporal overlap (`mean_niche_overlap_index` $\sim$ 0.09) and high niche similarity (`average_MVNH_dissimilarity` $\sim$ 0.03) suggests temporal niche partitioning, where species occupy similar sites at different times (Albrecht & Gotelli, 2001; Lear et al., 2021).

Future Directions

We will apply an approximate Bayesian computation framework to infer community assembly processes from long-term phytoplankton monitoring data. This project will depend on `MetaCommunityMetrics.jl` to calculate summary statistics in its pipeline.

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